

# HIRAM ZUNIGA

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## SUMMARY

Computer vision and machine learning engineer (CS '26). Author of 5 peer-reviewed papers and manuscripts, including 2 oral presentations at SPIE Optics + Photonics 2026. Podium finishes at MIT iQuHACK, Yale YQuantum, and NASA Space Apps. Hands-on experience shipping ML systems and event-driven data pipelines on AWS.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Swift

**ML / CV:** PyTorch, scikit-learn, OpenCV, YOLO, vision-language models (VLMs), CNNs, RAG, LLM fine-tuning, SentenceTransformers, UMAP, Optuna, Hugging Face Transformers, Ollama

**Cloud / Data:** AWS (Lambda, Step Functions, EventBridge, SQS, SNS, S3, DynamoDB, RDS, SageMaker, Bedrock), REST APIs, Power BI, Streamlit

**Tools:** Git/GitHub, unit testing, Linux, LaTeX

## EXPERIENCE

### Research Assistant – Computer Vision

Oct 2025 – Present

*CETYS University*

*Tijuana, MX*

- Fine-tuned a vision-language model and built a retrieval-augmented (RAG) pipeline for leukemia cell classification across multiple microscopy datasets; accepted as an **oral presentation at SPIE Optics + Photonics 2026**.
- Benchmarked CNN architectures for Alzheimer's disease detection in single-view MRI scans; **first-author oral presentation at SPIE Optics + Photonics 2026**.
- Developing a socially assistive human-following robot (deep learning, sensor fusion, CNN-based identity verification, LLM conversational interface); **first-author paper accepted at KES 2026**.
- Implemented and evaluated a Quantum SVM for Type-2 diabetes classification; first-author chapter in revision at *Advances in Computers* (Elsevier, Q1 book series).

### Machine Learning Capstone – Augmented AI for Thematic Analysis

Aug 2025 – Present

*California State University, Los Angeles*

*Remote*

- Built an end-to-end NLP pipeline (SentenceTransformers embeddings, UMAP reduction, clustering, Optuna hyperparameter optimization) that turns raw interviews/reviews into a prioritized theme map.
- Designed LLM-assisted labeling and summarization with controlled granularity via a hierarchical label tree.

### Data Analyst

Mar 2025 – Sep 2025

*I3PL (Cross-Border Logistics Solutions)*

*Tijuana, MX*

- Cut manual reporting work by 80% by automating daily multi-API data pipelines (Python, scheduled flows) publishing company-wide reports to SharePoint.
- Reduced container dwell time (>2 hours per load) through live milestone tracking and proactive alerts, preventing avoidable charges; owned end-to-end Power BI development.
- Replaced error-prone spreadsheets with a validated Power Apps capture system used across operations.

### AWS Mentorship – Message Router

Aug 2025 – Present

*Amazon Web Services mentorship program*

*San Diego, CA*

- Built an event-driven message-routing pipeline (EventBridge, SNS, SQS, Lambda, Step Functions, RDS) and a RAG chatbot on AWS; **demoed to AWS engineers at the Amazon San Diego office**.

## PUBLICATIONS (5 total – 3 first-author; full list: [hirambigdata.com/publications](http://hirambigdata.com/publications))

2 oral presentations at **SPIE Optics + Photonics 2026** (Applications of Machine Learning): Alzheimer's MRI CNN benchmark (first author) and retrieval-augmented VLMs for leukemia classification. First-author paper at **KES 2026** (*Procedia Computer Science*): distributed human-following robot. First-author chapter in revision, *Advances in Computers* (Elsevier): IoT + Quantum SVM for diabetes classification. Co-authored paper under review at **SPIE Security + Defence 2026**: LiDAR-based indoor robot navigation.

## HONORS & AWARDS

**3rd place**, MIT iQuHACK 2026 (Quantum Rings track) – ML prediction of quantum-algorithm runtime/fidelity · **3rd place**, Yale YQuantum 2026 – QAOA for portfolio optimization · **Regional 1st place & Global Nominee**, NASA International Space Apps Challenge 2025 – ExoVision, ML for exoplanet detection · **2nd place**, GEAR Web3 Regional Hackathon 2023 (USD \$1,000 prize) · NVIDIA Certificate: Getting Started with Deep Learning

## EDUCATION

### CETYS University

2022 – 2026

*B.S. Computer Science Engineering*

*Tijuana, MX*

- Coursework: Machine Learning, Deep Learning, Computer Vision, Parallel Computing, Evolutionary Computation, Data Structures, Algorithm Analysis, Probability & Statistics, Databases.
- Winter program in Big Data Analysis, Yonsei University, Seoul (2023–2024). Languages: English (TOEFL iBT 100), Spanish (native).